

Arjuna JEE 2025

Maths

DPP: 1

Trigonometric Equations

Q1 One of the principal solutions of $\sqrt{3} \sec x = -2$ is equal to

- (A) $\frac{2\pi}{3}$
 (B) $\frac{\pi}{6}$
 (C) $\frac{5\pi}{6}$
 (D) $\frac{\pi}{3}$

Q2 Find the principal value of the trigonometric function of $\cos x = \frac{\sqrt{3}}{2}$.

- (A) $\frac{\pi}{4}, \frac{5\pi}{4}$
 (B) $\frac{\pi}{4}, \frac{3\pi}{4}$
 (C) $\frac{\pi}{3}, \frac{2\pi}{3}$
 (D) $\frac{\pi}{6}, \frac{11\pi}{6}$

Q3 Find the principal solutions of the equation $\tan x = -\frac{1}{\sqrt{3}}$

- (A) $\frac{5\pi}{6}, \frac{11\pi}{6}$
 (B) $\frac{\pi}{6}, \frac{11\pi}{6}$
 (C) $\frac{7\pi}{6}$
 (D) None of these

Q4 Find the principal solution for $\tan x = -1$.

- (A) $\frac{2\pi}{3}, \frac{3\pi}{4}$
 (B) $\frac{3\pi}{4}, \frac{7\pi}{4}$
 (C) $\frac{\pi}{4}, \frac{5\pi}{4}$
 (D) None of these

Q5 Find the principal solution of $\cos x = -\frac{1}{2}$

- (A) $\frac{2\pi}{3}$
 (B) $\frac{\pi}{3}$
 (C) $\frac{\pi}{4}$

(D) $\frac{\pi}{6}$

Q6 $\cot \theta + \tan \theta = 2 \operatorname{cosec} \theta$, then find principal value of θ .

- (A) $\frac{\pi}{3}$ and $\frac{5\pi}{3}$
 (B) 0 and $\frac{2\pi}{3}$
 (C) $\frac{\pi}{3}$ and $\frac{\pi}{6}$
 (D) $\frac{\pi}{6}$ and $\frac{2\pi}{3}$

Q7 If $\cot \theta - \tan \theta = 2$, then find principal solution of θ .

- (A) $\frac{\pi}{4}, \frac{5\pi}{4}$
 (B) $\frac{\pi}{8}, \frac{5\pi}{8}$
 (C) $-\frac{\pi}{4}, -\frac{5\pi}{4}$
 (D) $-\frac{\pi}{8}, \frac{5\pi}{8}$

Q8 If $\cos \theta = -\frac{1}{2}$ and $0 < \theta < 360^\circ$ then the solution are

- (A) $\theta = 60^\circ, 240^\circ$
 (B) $\theta = 120^\circ, 240^\circ$
 (C) $\theta = 120^\circ, 210^\circ$
 (D) $\theta = 120^\circ, 300^\circ$

Q9 The number of value of θ in $[0, 2\pi]$ satisfying the equation $2 \sin^2 \theta = 4 + 3 \cos \theta$

- (A) 0
 (B) 1
 (C) 2
 (D) 43

Q10 The number of solutions of the equation $3 \sin^2 x - 7 \sin x + 2 = 0$ in the interval $[0, 5\pi]$ is

- (A) 0
 (B) 5
 (C) 6
 (D) 10



Answer Key

Q1 (C)

Q2 (D)

Q3 (A)

Q4 (B)

Q5 (A)

Q6 (A)

Q7 (B)

Q8 (B)

Q9 (A)

Q10 (C)

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